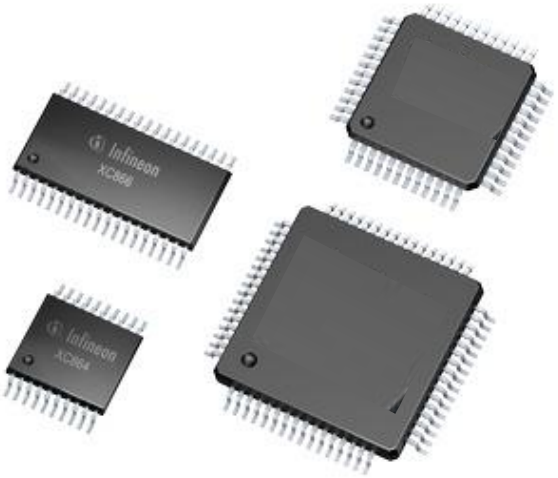
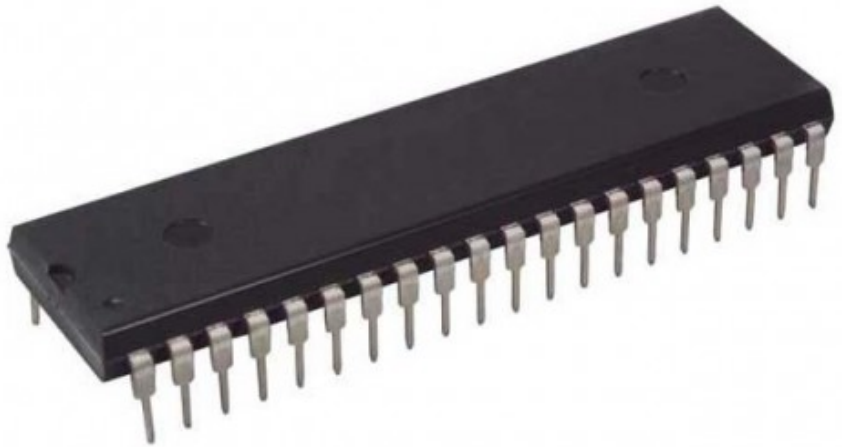
8051 Microcontroller





Assembler directives:

- ORG
- END
- EQU
- DB

ORG (origin)

- The ORG directive is used to indicate the beginning of the address
- The number that comes after ORG can be either in hex and decimal
- If the number is not followed by H, it is decimal and the assembler will convert it to hex

END

- This indicates to the assembler the end of the source (asm) file
- The END directive is the last line of an 8051 program
- Mean that in the code anything after the END directive is ignored by the assembler

EQU (equate)

- This is used to define a constant without occupying a memory location
- The EQU directive does not set aside storage for a data item but associates a constant value with a data label
- When the label appears in the program, its constant value will be substituted for the label
- Assume that there is a constant used in many different places in the program, and the programmer wants to change its value throughout
- By the use of EQU, one can change it once and the assembler will change all of its occurrences

```
COUNT   EQU   25
... ....
MOV   R3,   #COUNT
```

DB directives

- The DB directive is the most widely used data directive in the assembler
- It is used to define the 8-bit data
- When DB is used to define data, the numbers can be in decimal, binary, hex, ASCII formats

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- When DB is used to define data, the numbers can be in decimal, binary, hex, ASCII formats

```
ORG 500H
DB 28                   ;DECIMAL (1C in Hex)
DB 00110101B            ;BINARY (35 in Hex)
DB 39H                   ;HEX

DB “2591”               ;ASCII NUMBERS

DB “My name is Joe”     ;ASCII CHARACTERS
```